

LEC34 - Taylor and Maclaurin Series

Warm Up Problem

The Maclaurin series of f is a Taylor series of f but specifically centered at $a = 0$

Example

For $f(x) = \ln(x+1)$

a) Find the 4th Maclaurin polynomial of f

$$= f(0) + f'(0)x + \frac{f''(0)x^2}{2!} + \frac{f'''(0)x^3}{3!} + \frac{f^{(4)}(0)x^4}{4!}$$

$$= \ln(1) + \frac{x}{1} - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4}$$

b) Find the Maclaurin series for f

$$\Rightarrow p_{\infty} = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{x^n}{n} = \ln(1+x)$$

c) Show the radius of convergence in (b) is $R=1$

$\Rightarrow$ Use the ratio test

$$\Rightarrow \lim_{n \rightarrow \infty} \left| \frac{x^{n+1}}{n+1} \cdot \frac{n}{x^n} \right| = \lim_{n \rightarrow \infty} \left| \frac{x \cdot n}{n+1} \right| = x$$

$\Rightarrow$ We need $|x| < 1$, so $-1 < x < 1$, so $R=1$

d) Use the 4th Maclaurin series to approximate $\ln(1.5)$. How accurate is the remainder?

$$\Rightarrow \left[x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} \right] \approx \ln(1+x)$$

$$\Rightarrow 0.5 - \frac{(0.5)^2}{2} + \frac{(0.5)^3}{3} - \frac{(0.5)^4}{4} = 0.401 \quad (\ln(1.5) \approx 0.40)$$

$\Rightarrow$ Accurate within the next term $\frac{0.5^5}{5} \approx 0.00625$ by the Alternating Series Remainder Theorem

e) Can the series in (b) be used to approximate $\ln(2.5)$? Why or why not?

$\Rightarrow$ No, since $f(1.5) \approx \ln(2.5)$ is not valid since $x = 1.5 \notin (-1, 1)$, so the series will not converge

Example

Suppose you want to approximate $\sin(1.6328)$ using a series. What would be the best centre to use if we wanted to use as few terms as possible

- a) $x=0$ b) $x = \frac{\pi}{3}$ c) $x = \frac{\pi}{2}$ d) $x = 1.6$

Key Definitions

1. Taylor series at a : $\sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n$

2. n^{th} Taylor polynomial at a : $f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \dots + \frac{f^{(n)}(a)}{n!}(x-a)^n$

3. Maclaurin series (Taylor series at 0): $\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n$

4. n^{th} Maclaurin polynomial: $f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \dots + \frac{f^{(n)}(0)}{n!}x^n$

• If a power series centered at a converges to a function f , then that series must be the Taylor series of f at a

• Taylor and Maclaurin series are a systematic method for finding a power series of a function

• Taylor polynomials are the partial sums of Taylor series

• Taylor/Maclaurin polynomials of f can be used to approximate $f(x)$ for suitable values of x

 n^{th} Taylor Polynomial at a

$$f(x) \approx p_n(x) = f(a) + f'(a)(x-a) + \frac{f''(a)(x-a)^2}{2!} + \dots + \frac{f^{(n)}(a)(x-a)^n}{n!}$$

Example

Suppose you want to use a Taylor polynomial to approximate $f(\beta)$

Example

A large asteroid is approaching Earth. Suppose we know

$$f(0) = 600\text{km}, f'(0) = -35\text{km/s}, f''(0) = -2\text{km/s}^2, f'''(0) = 1\text{km/s}^3$$

How would we use to calculate the location of the asteroid after 3s?

$\Rightarrow$ 3rd degree Taylor polynomial, where $a=0, x=3$